DA3-SLAM: Pose-Conditioned Dense RGB SLAM with a Depth-Anything-3 Backbone on the $\mathrm{SL}(4)$ Manifold

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***Abstract*— DA3-SLAM addresses two questions about feed-forward visual-geometry SLAM built on the VGGT-SLAM submap / $\mathrm{SL}(4)$ / SALAD scaffold. First, does a stronger foundation-model backbone improve trajectory and depth quality? Through a thin adapter layer that leaves the rest of the pipeline intact, we replace VGGT with Depth-Anything-3 (DA3), a 1.4 B-parameter visual-geometry foundation model. With the giant metric-depth DA3 variant, DA3-SLAM reduces mean ATE RMSE from 11.48 m to 2.43 m and Eigen-crop AbsRel from 0.069 to 0.041 on the 13-drive KITTI validation split. Second, can a SLAM-refined trajectory in turn improve the network's depth prediction, and thereby further improve the trajectory? We introduce a pose-conditioned refinement pass that re-injects the optimised trajectory into DA3 to re-predict depth. On KITTI the answer is negative, for two reasons: DA3 does not self-correct from residual pose error in the input trajectory, and the $\mathrm{SL}(4)$ graph contains no depth factors, so pass-2 depth influences the optimiser only through a single inter-submap scale that pass-1 has already saturated. A 17-configuration ablation attributes the first-axis gain to the backbone swap and to backbone scaling. Code and the full ablation log are released with the paper. The code is available at `https://github.com/ZhongYeah1/DA3-SLAM`.**

I. Introduction

Feed-forward visual-geometry transformers such as DUSt3R [1], MASt3R [2], VGGT [3], and most recently Depth-Anything-3 (DA3) [4] predict dense depth, per-pixel confidence, and calibrated camera poses from a handful of images in a single forward pass, without explicit bundle adjustment. As SLAM front-ends, they enable a dense, feed-forward, per-submap-batched pipeline that contrasts with the sparse-feature back-end of ORB-SLAM3 [5] and with the iterative-BA loop of DROID-SLAM [6]. VGGT-SLAM [7] is the first system in this family: it runs VGGT per 17-frame submap, fuses the submaps through an $\mathrm{SL}(4)$ factor graph (which captures per-submap scale together with rotation and translation), and performs SALAD-based [8] loop closure.

This paper asks two questions around this design point:

Q1. Can a stronger foundation-model backbone (Depth-Anything-3 [4]) improve trajectory and depth quality on the same submap / $\mathrm{SL}(4)$ / SALAD scaffold?

Q2. VGGT-SLAM [7] demonstrates that the SLAM back-end can refine a neural network's initial pose estimates. The reverse direction remains open: can a SLAM-refined trajectory in turn improve the network's depth prediction, and does the improved depth further improve the trajectory? In other words, can the feed-forward network and the SLAM optimiser be mutually beneficial?

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For Q1, we answer in the affirmative. With the giant metric-depth variant of DA3 (1.40 B parameters), DA3-SLAM improves over VGGT-SLAM on the 13-drive KITTI validation split (mean ATE 2.43 m vs. 11.48 m; AbsRel 0.041 vs. 0.069), and the margin grows monotonically with backbone size.

For Q2, the answer on KITTI is negative, for two reasons:

- **DA3 does not self-correct from a refined pose input.** When the SLAM-optimised trajectory is supplied as input, the predicted depth does not improve; it is in fact marginally degraded relative to pass-1. The refined trajectory still contains residual pose error, and this residual misleads the depth head.
- **Depth influences the trajectory only through a single scalar.** In the $\mathrm{SL}(4)$ graph, the sole mechanism by which pass-2 depth can improve the trajectory is through an updated inter-submap scale s. Pass-1 already estimates s tightly, leaving little residual signal for pass-2 to exploit (Sec. VI-E).

Contributions:

- **A thin adapter** that ports the VGGT-SLAM submap / $\mathrm{SL}(4)$ / SALAD stack to DA3 without modifying either upstream codebase. Only three interface layers are affected: extrinsic and depth-confidence formats, camera-space unprojection, and a median-aggregated attention verifier for loop closure.
- **A pose-conditioned refinement pass** (Phase B). After pass-1, the optimised trajectory is re-injected into DA3 through its pose-conditioning module to re-predict depth. We compare three graph re-optimisation strategies: bounded-clip rebuild, in-place factor replacement, and skip.
- **A 17-configuration ablation** on the 13-drive KITTI validation split across seven axes: backbone choice, backbone size, pass-2 refinement, re-optimisation variant, scale estimator, chunk size and overlap, and keyframe disparity. Backbone size dominates every other axis by $4\times$ or more; every refinement variant either ties or regresses the baseline ATE.
- **A Pareto improvement over VGGT-SLAM on KITTI**: -79% ATE and -41% AbsRel with the giant metric-depth backbone. The largest per-drive improvements occur on long drives without loop closure: $36.8 \rightarrow 2.4$ m on drive 0023, $53.2 \rightarrow 8.4$ m on drive 0036, and $22.0 \rightarrow 10.2$ m on drive 0047.

II. Related Work

Feed-forward visual-geometry transformers. DUSt3R [1] predicts dense point maps and pixel-wise confidence from image pairs in a single pass; MASt3R [2] adds metric scale and a matching head. VGGT [3] generalizes to arbitrary numbers of input views with a transformer that jointly produces camera parameters, depth maps, point maps, and 3D point tracks. Depth-Anything-3 [4] builds on the Depth Anything lineage [9], [10] by separating the visual backbone (vanilla DINOv2 [11]) from a *DualDPT* [12] depth-ray decoder and a pose-conditioning head, enabling cross-view consistency via token concatenation in the backbone rather than through a separate aggregator.

Dense feed-forward SLAM. DROID-SLAM [6] is the canonical recurrent-deep-BA monocular SLAM. More recent systems replace the per-iteration update with a single large forward pass. VGGT-SLAM [7] is the closest system to ours: it runs VGGT per submap, converts per-submap scale ambiguity into an $\mathrm{SL}(4)$ manifold parameterisation, fuses submaps through pairwise between-submap factors, and performs SALAD-based loop closure. We adopt its graph and loop-closure scaffold verbatim and ablate only the backbone and the refinement pass.

Factor-graph back-ends. We use the GTSAM [13], [14] Levenberg–Marquardt optimizer with the SL(4) retraction provided by the VGGT-SLAM authors; iSAM2 [15] is a natural drop-in that we do not explore here because batching per submap already keeps per-step cost bounded.

Benchmarks. We report KITTI [16] validation metrics (ATE via Umeyama Sim(3) alignment [17] using the *evo* toolkit [18], and depth via the Eigen crop [19] with median alignment), together with qualitative results on a TUM RGB-D [20]-style indoor sequence (*office_loop*). We reference but do not re-run the classical ORB-SLAM3 baseline [5]; the closest comparable dense feed-forward peer is VGGT-SLAM.

III. Problem Formulation

Given a sequence of RGB frames $\{I_t\}_{t=1}^{T}$ from an uncalibrated monocular camera, we seek (i) a trajectory $\{T_t\}_{t=1}^{T} \subset \mathrm{SE}(3)$ and (ii) a dense reconstruction $\mathcal{P} \subset \mathbb{R}^3$ that is globally consistent under loop closures. The system is allowed to select a sparse set of keyframes $\{t_k\}_{k=1}^{K}$ and operate in submap batches; the design target is *per-submap batched feed-forward inference* so that no per-pixel online bundle adjustment is required.

Challenge 1: Per-submap projective ambiguity. Each submap is produced by an independent forward pass of the feed-forward network, and each pass normalises its output in its own internal frame [1], [4]. Two neighbouring submaps therefore live in different reconstruction frames that must be reconciled before fusion. Without calibration or motion priors, a classical result in multi-view geometry establishes that any two such reconstructions of the same scene are related only up to a general 15-DoF projective transformation, so neither a pure scale factor nor a similarity transform ($\mathrm{Sim}(3)$, 7 DoF) can absorb the full ambiguity. Following VGGT-SLAM [7], we therefore lift the inter-submap transform to the $\mathrm{SL}(4)$ manifold — the 15-dimensional group of $4{\times}4$ determinant-one homographies — whose dimension matches the projective ambiguity exactly.

Challenge 2: Long-range error accumulation. On driving sequences without loop closures, per-submap pose error accumulates along the trajectory. This constitutes the dominant failure mode on KITTI, and the axis along which larger backbones yield the greatest improvement (Sec. VI).

Challenge 3: Closing the loop on depth. VGGT accepts only images as input. DA3 additionally accepts extrinsics and intrinsics, injecting them into the DINOv2 backbone via cross-attention [4]. This capability, absent from VGGT, motivates the question of whether depth can be improved when a refined trajectory is available — the starting point for Phase B.

IV. Approach

The pipeline (Fig. 1) runs in two stages. **Phase A** is a single-pass, per-submap incremental SLAM loop inherited from VGGT-SLAM: DA3 is called once per submap, and each submap is fused into an $\mathrm{SL}(4)$ factor graph with SALAD-based loop closure, producing a baseline trajectory and dense geometry. **Phase B** is the pose-conditioned refinement contributed in this work: the Phase-A trajectory is re-injected into DA3 as explicit pose input, the resulting pass-2 depth is scale-aligned and substituted into each submap, and the graph is optionally re-optimised against the refined geometry.

A. Adapter layer

We leave both upstream codebases unmodified. A thin adapter translates between DA3's output (depth, confidence, world-to-camera extrinsics, pinhole intrinsics) and the data types expected by the solver (camera-space point clouds, $4{\times}4$ w2c poses, and intrinsic blocks embedded in a $4{\times}4$ matrix). The only geometrically non-trivial step is unprojection: we unproject to camera coordinates rather than to world coordinates, so that the $\mathrm{SL}(4)$ node remains the sole carrier of the world transform and downstream graph updates automatically propagate to every point cloud without re-unprojection. The SLAM core, loop-closure detector, graph, optimiser, and visualiser are otherwise identical to VGGT-SLAM, making our KITTI comparison a controlled backbone swap.

B. Phase A — Per-submap incremental SLAM

Keyframe selection. Lucas–Kanade [21] sparse optical flow tracks Shi–Tomasi [22] corners forward from the last keyframe, and a frame is promoted to a keyframe once the mean pixel displacement exceeds a disparity threshold $d_{\min}$. We ablate $d_{\min} \in \{10, 15, 25\}$ (Sec. VI).

DA3 inference (pass 1). Each submap consists of C consecutive keyframes, sharing a user-specified overlap with the next submap. DA3 is called once per submap and returns, for each frame, a depth map, a per-pixel confidence map, a

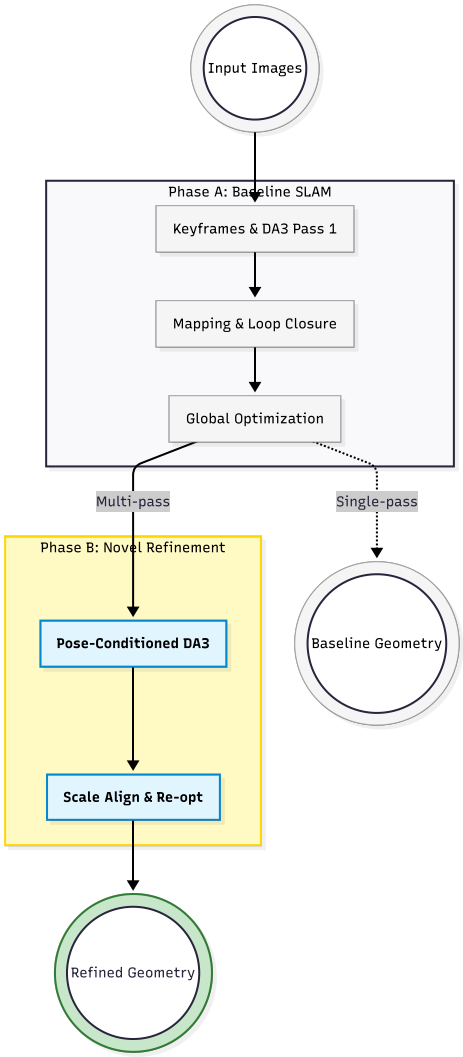

Fig. 1: **DA3-SLAM pipeline.** The system runs in two stages. **Phase A** (single-pass, baseline SLAM) takes the input images, selects keyframes and sends them through a first DA3 forward pass, performs mapping and SALAD-based loop closure, and applies global $\mathrm{SL}(4)$ optimisation to yield a baseline trajectory and geometry. **Phase B** (multi-pass, our contribution) then feeds the optimised trajectory back into DA3 as explicit pose conditioning; the resulting pass-2 depth is aligned in scale, substituted into the affected submaps, and the $\mathrm{SL}(4)$ graph is optionally re-optimised, yielding a refined trajectory and geometry.

world-to-camera extrinsic, and a pinhole intrinsic. We force DA3's `ref_view_strategy`="first" so that frame 0 has the identity extrinsic; this preserves the SLAM convention of a canonical reference frame, which DA3's default view-ordering heuristic would otherwise permute.

Inter-submap scale estimation. Because DA3 normalises depth per forward pass, the point clouds of consecutive submaps differ by an unknown scalar factor s that must be recovered before they can be fused. Using the overlap frames, we implement three pluggable estimators: MEDIAN, the median ratio of pairwise point distances; DEPTH-RANSAC, RANSAC on the linear fit $d_{\text{prev}} = s \cdot d_{\text{curr}}$ with a confidence-pruning mask; and DEPTH-WEIGHTED, a confidence-weighted quantile regression with weights $(\text{conf}_{\text{prev}} \cdot \text{conf}_{\text{curr}})^2$. Each estimator also returns a quality score $q \in [0, 1]$ from its inlier statistics; if $q < 0.3$ the system falls back to MEDIAN.

$\mathrm{SL}(4)$ **factor graph.** Each submap contributes one $\mathrm{SL}(4)$ node H_k (a $4{\times}4$ determinant-one homography), and each keyframe inside the submap is stored as a known offset from this node. Three factor families connect the nodes:

- A *prior* on submap 0 with $\sigma = 10^{-6}$, anchoring the global frame.
- *Intra-submap* factors between consecutive keyframes, $H^{\text{intra}}_{i\to i+1} = w2c_i \cdot w2c_{i+1}^{-1}$, with $\sigma = 0.05$; these encode DA3's per-frame poses.
- *Inter-submap* factors that embed the estimated scale,

$$H^{\text{inter}}_{k,k+1} = K_k^{-1} \cdot K_{k+1} \cdot \mathrm{diag}(s, s, s, 1), \qquad (1)$$

  where K_k is the intrinsic of submap k's anchor frame embedded in a $4{\times}4$ block. When consecutive submaps share the same intrinsic (the usual case on KITTI), this reduces to a pure scale homography. Consequently, s is the only pass-2-sensitive quantity in the graph — a fact that explains the pass-2 ATE results in Sec. VI.

GTSAM Levenberg–Marquardt [13] is run on the $\mathrm{SL}(4)$ manifold after every submap, with a guard that catches the occasional singular retraction produced by a degenerate scale estimate and retains the current iterate.

Loop closure: SALAD + attention verification. Each keyframe is passed through SALAD [8] to produce a global descriptor for place recognition. Candidate pairs below an ℓ_2 threshold are verified by re-running the pair through DA3 and computing a *cross-attention ratio* on the keys and queries of a late DINOv2 block (extracted via a forward hook): the fraction of inter-frame attention mass relative to the peak self-attention response. We adapt the VGGT-SLAM formulation in two respects. First, the CLS-token offset is reduced from 5 to 1, because DA3's DINOv2 uses a single special token rather than VGGT's five. Second, per-head ratios are aggregated by *median* rather than by the mean of the top quartile: DA3's query–key normalisation and rotary position embeddings produce heavy-tailed per-head distributions for which the mean is unstable. A cross-attention ratio of 0.50 admits the closure; accepted loop closures are wired into the graph as a two-frame submap connected to both endpoints.

C. Phase B — Pose-conditioned refinement

Once Phase A converges, the optimised trajectory is re-injected into DA3 to produce a second-pass geometry. The procedure consists of five steps: (a)–(b) produce pass-2 depth conditioned on the refined trajectory; (c) brings pass-2 into the same internal scale as pass-1; (d) substitutes the scaled pass-2 geometry into each submap; and (e) optionally updates the $\mathrm{SL}(4)$ graph to reflect the substituted geometry.

(a) Per-frame pose extraction. The optimiser stores one $\mathrm{SL}(4)$ node H_k per submap rather than per-frame poses, so we first recover the per-frame extrinsics that will condition DA3. For frame i in submap k, we build the composite projection $P_i = K_i \cdot H_k^{-1}$ and extract camera parameters by RQ decomposition,

$$\big(K,\ R_{c2w},\ t_{c2w}\big) \ = \ \mathrm{rq}(P_i). \tag{2}$$

Because RQ returns a *camera-to-world* pose but DA3 expects *world-to-camera*, we invert by transposing the rotation and negating the transformed translation,

$$R_{w2c} = R_{c2w}^{\top}, \qquad t_{w2c} = -R_{c2w}^{\top}\, t_{c2w}. \tag{3}$$

(b) Pose-conditioned forward pass. The extracted extrinsics are first anchored to the identity at frame 0 and then rescaled so that their translations have unit median norm, placing the pose cloud near the radius on which DA3's pose encoder was trained. DA3 then encodes each extrinsic as a 9-dimensional token (translation, unit quaternion, and two field-of-view scalars) and injects it into a mid-layer of the DINOv2 backbone via cross-attention, producing a trajectory-conditioned depth and point-cloud prediction.

We explicitly disable DA3's built-in post-prediction alignment step, which performs a $\mathrm{Sim}(3)$ alignment between its predicted and supplied poses and rescales the output depth by the recovered factor. At this point, the supplied poses are already in graph-world scale while DA3's output depth is in its per-call internal scale, so a $\mathrm{Sim}(3)$ fit here would introduce a persistent scale mismatch rather than remove one. Scale alignment is instead performed separately in step (c).

(c) Scale alignment. Because DA3 is pose-conditioned, its pass-2 rotations and intrinsics already agree with pass-1; only translation magnitudes (and the corresponding depth scale) require reconciliation. We use a closed-form estimator on translation norms,

$$s^{\star} \ = \ \mathrm{clip}\left(\frac{\mathrm{mean}_{i\geq 1}\, \|t_i^{(1)}\|_2}{\mathrm{mean}_{i\geq 1}\, \|t_i^{(2)}\|_2},\ 10^{-2},\ 10^{2}\right), \tag{4}$$

where $t_i^{(1)}$ and $t_i^{(2)}$ are the pass-1 and pass-2 camera-to-world translations. Frame 0 is excluded from the mean because the identity anchoring in step (b) forces $t_0^{(\cdot)} = \mathbf{0}$. The factor $s^{\star}$ is applied jointly to pass-2 depth and pass-2 translations, keeping the two scale-linear quantities consistent with rotation and intrinsics, which are scale-invariant.

(d) Submap substitution. All per-submap pass-1 quantities — per-frame poses, intrinsic blocks, point clouds, colours, confidence, and confidence masks — are overwritten with their pass-2 counterparts after rescaling by $s^{\star}$. This happens before step (e), so every graph-reoptimisation variant (including NONE) evaluates on pass-2 geometry; only the output trajectory differs across variants.

(e) Graph re-optimisation. Three variants are selected by `--reopt_method`:

- **BOUNDED-CLIP** (default). The $\mathrm{SL}(4)$ graph is rebuilt from the recorded edge log. For each non-loop-closure edge, the inter-submap scale is recomputed from pass-2 geometry and then clipped to $s_{\mathrm{pass1}} \cdot [1-\alpha,\ 1+\alpha]$ with $\alpha = 0.05$, after which LM is re-run from a cold start. The clip limits the influence of pass-2 scale outliers at the cost of a heuristic parameter.
- **INPLACE.** The pass-1 graph is retained; only its inter-submap factors are replaced with pass-2-derived measurements, and LM is warm-started from the pass-1 solution. Prior, intra-submap, and loop-closure factors keep their pass-1 measurements.
- **NONE.** The graph is not re-optimised, so the trajectory is byte-identical to Phase A while the submap geometry still reflects pass-2. On KITTI this variant ties or improves upon the other two at every backbone size (Sec. VI).

V. Experiments

Dataset. We evaluate on the 13-drive KITTI validation split [16], resolution 1242×375 at 10 Hz. The drives span urban, residential, and country-road scenes of length 15–460 m, and none of them contain loop closures, so KITTI stresses long-range pose accumulation rather than SALAD retrieval. We also check loop closure qualitatively on an indoor *office_loop* sequence, where the attention verifier makes two correct accept/reject decisions and the loop gap relative to GT decreases from 37% to 6.5% of the trajectory length.

Metrics. For trajectories we report ATE RMSE after Umeyama $\mathrm{Sim}(3)$ alignment to the ground-truth trajectory [17], [18]. For depth we report AbsRel, RMSE, and $\delta < 1.25$ under the standard Eigen crop [19], with per-frame median scale alignment to the KITTI sparse-LiDAR projection. All means are unweighted averages over the 13 drives.

Baseline. Our direct peer is VGGT-SLAM [7] with its default settings (chunk size 32, overlap 8, minimum disparity 15 pixels). Classical feature-based baselines such as ORB-SLAM3 [5] and DROID-SLAM [6] differ in paradigm: ORB-SLAM3 is sparse-feature, and DROID-SLAM is iterative rather than per-submap batched. We therefore focus the experimental budget on same-paradigm ablations.

Implementation. All runs use a single NVIDIA L40S (48 GB) and a development build of GTSAM that provides the $\mathrm{SL}(4)$ retraction. Each configuration is run across all 13 drives and produces a TUM trajectory together with per-frame depth; metrics are computed by the *evo* toolkit and a depth scorer that respects the Eigen crop.

VI. Results and Analysis

We have uploaded a video to explain the results in more detail: https://youtu.be/1Zi687jzQ4Y.

A. Headline comparison with VGGT-SLAM

Table I reports KITTI means over 13 drives with the strongest DA3 backbone (the giant metric-depth variant, 1.40 B parameters) against the VGGT-SLAM baseline. Without any refinement, DA3-SLAM reaches a mean ATE of

TABLE I: **Headline KITTI validation result** (mean over 13 drives). Best per column in **bold**. With the giant metric-depth backbone, DA3-SLAM improves on VGGT-SLAM in both ATE and AbsRel simultaneously.

Method	ATE[m]↓	AbsRel↓	RMSE_d ↓	$\delta_{1.25}$ ↑
VGGT-SLAM [7]	11.48	0.069	4.33	0.938
DA3-SLAM (small)	4.64	0.084	4.34	0.915
DA3-SLAM (giant)	3.92	0.050	2.91	0.969
DA3-SLAM (giant, metric)	**2.43**	**0.041**	**2.61**	**0.978**

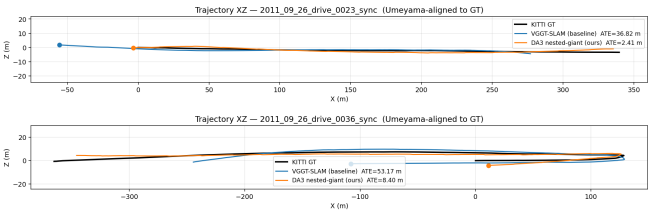

Fig. 2: **Trajectory comparison, KITTI drives 0023 and 0036** (Umeyama-aligned to GT). VGGT-SLAM's per-submap scale ambiguity compounds over the drive, deflecting the trajectory from the initial direction (blue); DA3-SLAM with the giant metric-depth backbone (orange) remains close to GT throughout. ATE drops $36.8 \rightarrow 2.4$ m on 0023 and $53.2 \rightarrow 8.4$ m on 0036.

2.43 m and AbsRel of 0.041, improving VGGT-SLAM on both axes simultaneously. The largest per-drive improvements line up with the drives on which VGGT-SLAM's relative-scale drift is largest: $15\times$ on drive 0023, $6\times$ on drive 0036, $2\times$ on drive 0047 (Fig. 2).

B. Backbone scaling (Axis 1)

Table I isolates the backbone axis by holding everything else constant. Scaling the backbone from the small variant to the giant variant to the giant metric-depth variant reduces ATE from 4.64 to 3.92 to 2.43 m and AbsRel from 0.084 to 0.050 to 0.041, monotonic in both metrics. Among the axes we sweep, backbone scale has by far the largest effect: it is what moves DA3-SLAM from *improving ATE but regressing AbsRel relative to VGGT-SLAM* at the small variant to improving both at the giant metric-depth variant.

C. Mechanism: metric-scale depth and sky regions

Fig. 3 visualises the depth-map difference that underlies both the AbsRel and the ATE improvement. VGGT-SLAM produces sharper depth edges but at a relative scale that varies across submaps, whereas the giant metric-depth DA3 variant produces metric-scale depth that is consistent across the entire sequence. Combined with the per-edge $\mathrm{SL}(4)$ scale estimator, near-metric depth causes the inter-submap scale factors to drift less, consistent with the near-zero trajectory bias observed in Fig. 2.

A closer inspection of Fig. 3 reveals a more specific failure mode that accounts for a substantial share of the gap. KITTI is an outdoor driving benchmark: every keyframe contains a large sky region, typically occupying 20–40% of the image. DA3 consistently places the sky at the far end of its predicted depth range, as expected for regions at effectively unbounded

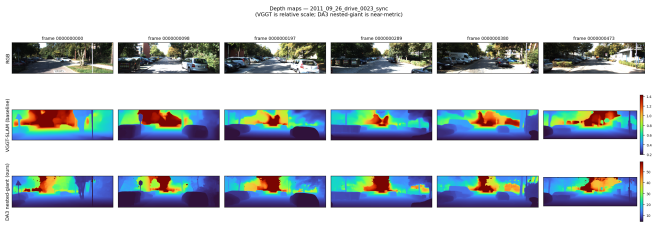

Fig. 3: **Depth-map comparison, KITTI drive 0023.** Six keyframes (columns) for RGB, VGGT-SLAM, and DA3-SLAM with the giant metric-depth backbone. VGGT's output is relative ($\sim [0.2, 2.7]$) and does not share a common scale across frames; note that patches of the sky receive short, finite depth values. The giant metric-depth DA3 variant produces near-metric depth ($\sim [5, 115]$ m) and consistently places the sky at its far clipping plane. These two effects — cross-frame scale consistency and correct sky geometry — together drive the -41% AbsRel reduction and a substantial share of the ATE improvement (Sec. VI-C).

distance. VGGT, in contrast, occasionally assigns short, finite depths to patches of the sky — a known failure mode of relative-depth models that are not explicitly supervised on open-sky geometry. Because our inter-submap scale estimator is derived from pairwise point-distance and depth-ratio statistics, these spurious short-depth sky pixels act as outliers that bias the estimated scale toward smaller values. The resulting error propagates through the $\mathrm{SL}(4)$ edge and compounds along the trajectory, which is precisely the failure mode dominating the long no-loop-closure drives on which the ATE gap is largest (Fig. 2, drives 0023 and 0036). Correct far-field sky geometry is therefore a disproportionate contributor to trajectory accuracy on outdoor driving data, and it is one of the properties most directly improved by a metric-depth foundation model.

D. Refinement pass (Axis 2)

Table II reports the Phase B ablation with the small backbone. Every refinement variant we evaluated either ties or regresses the pass-1 ATE of 4.64 m: NONE ties exactly by construction (the graph values are unchanged), BOUNDED-CLIP costs $+25\%$, and INPLACE costs $+65\%$. Depth metrics are statistically flat, with AbsRel moving only $0.084 \rightarrow 0.085$ (within per-drive noise). The same pattern holds at the giant and giant metric-depth backbones: REFINE/NONE matches the baseline exactly, and REFINE/INPLACE adds $+5\%$ ATE at the giant metric-depth backbone (Table I).

E. Why refinement is ATE-neutral on KITTI

The $\mathrm{SL}(4)$ graph contains a single 4×4 node per submap and admits no per-pixel depth, reprojection, or bundle-adjustment residuals. Pass-2 depth therefore enters the optimiser only through the scalar s embedded in each inter-submap edge. On KITTI drives, at a minimum disparity of 15 pixels, consecutive keyframes exhibit stable overlap geometry, and the pass-1 MEDIAN estimator already fits s to within a small tolerance; pass-2 perturbs s by sub-percent amounts that LM treats as noise. The null result therefore

TABLE II: **Phase B ablation** (small backbone, 13 KITTI drives). Every re-optimisation variant ties or regresses the baseline ATE; depth is flat. The $\mathrm{SL}(4)$ graph has no depth factors, so pass-2 depth has no mechanical path into the ATE cost (Sec. VI-E).

Configuration	ATE↓	AbsRel↓	RMSE_d ↓	δ↑
Baseline (pass 1)	**4.64**	**0.084**	4.34	**0.915**
Refine, NONE	**4.64**	0.085	**4.29**	0.913
Refine, BOUNDED-CLIP	5.80 (+25%)	0.085	**4.29**	0.913
Refine, INPLACE	7.67 (+65%)	0.085	**4.29**	0.913
Large-chunk refine	9.00 (+94%)	0.085	4.32	0.912

TABLE III: **Ablation axes 3–5** (small backbone, 13 KITTI drives). Baseline: MEDIAN scale, chunk size 17 with overlap 1, and minimum disparity of 15 pixels. None of the axes beats the baseline on ATE. Depth metrics are essentially invariant across (3) and (4), since scale and chunk are same-frame rescalings; only keyframe density moves AbsRel measurably.

Axis	Config	ATE↓	AbsRel↓
3. Scale	MEDIAN (*baseline*)	**4.64**	0.084
	DEPTH-RANSAC	4.73	0.084
	DEPTH-WEIGHTED	7.70 (+66%)	0.084
4. Chunk×Ov	17×1 (*baseline*)	**4.64**	**0.084**
	32×4	5.00	0.086
	60×8	16.92 (+265%)	0.087
5. Min. disp.	10	6.13	**0.0836**
	15 (*baseline*)	**4.64**	0.0839
	25	5.13	0.0860

combines two factors: (i) the graph provides no mechanism for depth beyond a single scalar per edge, and (ii) on KITTI, pass-1 has already exhausted the information that scalar can carry. Sec. VII discusses the benchmarks on which (ii) is expected to relax.

F. Remaining axes

Table III summarises the remaining four axes. Each is a mechanism we implemented that did not improve KITTI; we report them because the negative results help narrow the next design iteration. On scale estimation (Axis 3), MEDIAN performs best; DEPTH-WEIGHTED regresses because KITTI sky and road pixels dominate the confidence mass and bias the weighted quantile. On chunk size (Axis 4), a chunk size of 17 is a local optimum, while a chunk size of 60 sharply degrades long drives without loop closure ($+265\%$ ATE). On keyframe density (Axis 5), a minimum disparity of 15 pixels is the sweet spot.

VII. Limitations and Future Work

A. Limitations

Single-benchmark evaluation. All quantitative results in this paper are obtained on a single dataset: the KITTI validation split, which is entirely outdoor and contains no loop closures. The extent to which the Q1 improvement and the Q2 null result generalise to indoor scenes, loopy sequences, or other outdoor distributions is not established.

The pose-conditioned refinement is an unrealised capability. Phase B is technically enabled by DA3's pose-input interface, yet it yields no ATE improvement on KITTI. Two obstacles render it latent in practice: DA3 does not correct its depth prediction in response to the residual pose error that remains after pass-1, and the $\mathrm{SL}(4)$ graph admits no per-pixel depth or reprojection residuals, so pass-2 depth reaches the optimiser only through a single scalar per edge.

No global bundle adjustment. The back-end optimises only inter-submap $\mathrm{SL}(4)$ transforms, with no pixel-level reprojection residuals and no joint refinement of poses and a sparse reconstruction. As a direct consequence, a large fraction of the depth information produced by DA3 never reaches the trajectory cost.

Offline-only throughput. With the giant metric-depth backbone the system runs at ~ 4 FPS on a single L40S; the small backbone reaches 12 FPS at reduced accuracy. Neither configuration meets the real-time budget of an onboard robotics stack.

Inherited pipeline constraints. We build on the original VGGT-SLAM's submap / $\mathrm{SL}(4)$ / SALAD scaffold and therefore inherit its known shortcomings, including the 15-DoF drift and planar-degeneracy modes associated with optimising on the full $\mathrm{SL}(4)$ manifold.

Post-hoc scale recovery. Metric scale is still obtained by $\mathrm{Sim}(3)$ alignment against the ground-truth trajectory rather than from the system itself; an IMU- or GPS-based front-end would be needed to remove this dependency.

B. Future Work

Benchmarking on diverse sequences. All numbers in this paper come from a single outdoor driving benchmark. Before any of our findings can be generalised, the same 17-configuration sweep should be repeated on qualitatively different data: indoor sequences such as TUM RGB-D [20] or ScanNet, where short baselines leave pass-1 with more scale ambiguity for the refinement pass to exploit, and longer outdoor sequences with explicit loop closures, on which the SALAD retrieval plus attention verification front-end can be stress-tested. Such a sweep will separate structural properties of the system — which should carry over — from KITTI-specific effects, such as the sky-region mechanism of Sec. VI-C.

Global bundle adjustment. The simplest route to making Phase B actually help is to give depth a direct path into the trajectory cost. A global bundle adjustment, jointly refining camera poses and a sparse 3D reconstruction against pixel-level reprojection residuals, would do exactly this. Such a stage would also address the broader "no depth factor" limitation above, and is the natural way to finally close the pose-depth-pose feedback loop that Q2 set out to test but that the current factor graph cannot express. An incremental back-end [15] would be needed to keep the additional factor count tractable.

Stronger backbones and stronger SLAM pipelines. The present system is essentially "DA3 plugged into the original VGGT-SLAM pipeline," and both ends can be improved

independently. On the backbone side, a streaming variant of DA3 can trade a small amount of accuracy for real-time latency, and future visual-geometry foundation models can be dropped in through the same thin adapter. On the SLAM side, more recent feed-forward back-ends abandon the full $\mathrm{SL}(4)$ parameterisation in favour of lower-dimensional, less drift-prone alternatives, and would remove the inherited constraint discussed above. Pairing a stronger backbone with a stronger back-end is the obvious path to pushing both ATE and throughput substantially further than the present system.

VIII. Conclusion

We presented DA3-SLAM, built by replacing the VGGT backbone of VGGT-SLAM with Depth-Anything-3 through a thin adapter, and adding a pose-conditioned refinement pass on top. On the 13-drive KITTI validation split, the backbone swap alone reduces mean ATE from 11.48 m to 2.43 m (-79%) and AbsRel from 0.069 to 0.041 (-41%), driven by metric-scale depth and correct far-field sky geometry. The refinement pass is ATE-neutral on KITTI: DA3 does not self-correct from residual pose error, and the $\mathrm{SL}(4)$ graph provides no path for depth into the trajectory cost beyond a single scalar per inter-submap edge. Backbone quality dominates every other axis we examined; closing the pose–depth–pose loop will require a factor graph that actually exposes depth to the optimiser. Code and the full ablation log are released with the paper.

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